GENERAL APTITUDE

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What is **Aptitude ?**

It is your natural ability to learn or excel in a certain area.

For example, you could have an **aptitude** for math and logic.

Key to **success**

1. Problem Recognition
2. Speed
3. Practice

Link for **English Basics**

- https://www.learngrammar.net/practice
- https://www.myenglishpages.com/english/exercises.php
- https://www.englisch-hilfen.de/en/exercises_list/alle_grammar.htm
- https://www.grammarbank.com/
- https://www.really-learn-english.com/english-grammar-exercises.html
- https://www.really-learn-english.com/english-reading-comprehension-text-and-exercises.html
- https://www.thefreshreads.com/unseen-passage-for-class-10/
- https://www.englishgrammar.org/exercises/

- Practice Synonyms and Antonyms regularly.
- Read Idioms and Phrases.
- Book - Word Power Made Easy by Norman Lewis
- Book - English Grammar by Wren and Martin

Basic **MATHS**

- Tables at least from 2-30
- Squares from 2-30
- Cubes from 2-30
- Prime numbers from 1-100
- Divisibility rules for 1-20

- Methods for typical multiplications & divisions
- Methods for finding HCF & LCM
- Methods for finding squares & square roots

Topic Wise Test Plan

TEST NAME	TOPICS
APTI 1	Numbers + LCM + HCF + Ages + Averages
APTI 2	Percentages + Allegations & Mixtures + Profit & Loss
APTI 3	Time & Work + Pipes & Cisterns + Chain Rule
APTI 4	Time & Distance + Trains + Boats + Interest
APTI 5	Clock + Calendar + Probability + Permutation Combination

Numbers

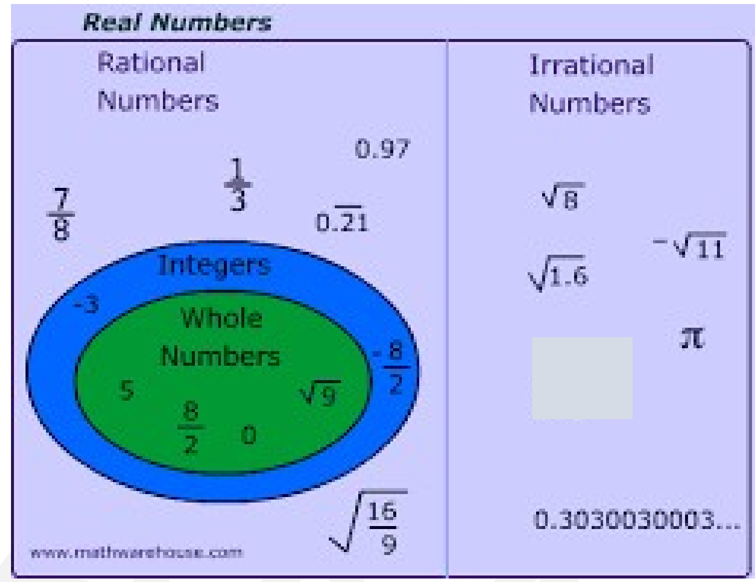
Real

Complex (a+bi)

A complex number is a number that can be expressed in the form a + bi, where a and b are real numbers, and i is a solution of the equation $x^2 = -1$. Because no real number satisfies this equation, i is called an imaginary number

Rational (-2, ¾, 2.25,)

Irrational (√2, 3.14159….)



Integers (∞….,-2,-1,0,1,2….∞)

Whole Nos (0,1,2,3.. ∞)

Negative Nos (∞ ..-2,-1)

Fractions (2/5, 4/3, 3 1/3)

Natural Nos (1,2,3… ∞)

Zero (0)

Proper Fractions (2/5, 3/7..)

Improper Fractions (4/3, 8/7..)

Mixed Fractions (3 ⅔, 4 ⅓..)

Prime Nos

Composite Nos (3 distinct factors)

What is the Difference Between Rational Numbers and Irrational Numbers?

Rational Numbers	**Irrational Numbers**
Numbers that can be expressed as a ratio of two numbers (p/q form) are termed as a rational number.	Numbers that cannot be expressed as a ratio of two numbers are termed as an irrational number.
Rational Number includes numbers, which are finite or are recurring in nature.	These consist of numbers, which are non-terminating and non-repeating in nature.
If a number is terminating number or repeating decimal, then it is rational. e.g: 1/2 = 0.5	If a number is non-terminating and non-repeating decimal, then it is irrational. e.g: 0.31545673...
Example:- 1/2, 3/4, 11/2, 0.45, 10, etc.	example:-Pi (π) = 3.14159...., Euler's Number (e) = (2.71828...), and √3, √2.

Basic **MATHEMATICAL** operations

- BODMAS
- B - Bracket ( ) , { }, [ ]
- O - Order
- D - Division
- M - Multiplication
- A - Addition
- S - Subtraction.

BASIC FORMULAE

- 1. $(a + b)^2 = a^2 + b^2 + 2ab$
- 2. $(a - b)^2 = a^2 + b^2 - 2ab$
- 3 . $(a + b)^2 - (a - b)^2 = 4ab$
- 4. . $(a + b)^2 + (a - b)^2 = 2 (a^2 + b^2)$
- 5. $(a^2 - b^2) = (a + b) (a - b)$
- 6. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2 (ab + bc + ca)$
- 7. $(a^3 + b^3) = (a + b) (a^2 - ab + b^2)$
- 8. $(a^3 - b^3) = (a - b) (a^2 + ab + b^2)$
- 9. $(a^3 + b^3 + c^3 - 3abc) = (a + b + c) (a^2 + b^2 + c^2 - ab - bc - ca)$
- 10. If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$

ADDITION

- SUM of 2 EVEN numbers is EVEN : 2 + 6=8
- SUM of 2 ODD numbers is EVEN : 7 +17=24
- SUM of ODD & EVEN numbers is ODD : 13 + 8=21

LAST DIGIT is same as last digit of sum of last digits

- E.g. For 431 + 632 + 233 + 539 + 845

  Last digit is the last digit of 1+2+3+9+5 = 20

  So last digit = 0

SUBTRACTION

- DIFF of 2 EVEN numbers is EVEN : $6 – 2 = 4$
- DIFF of 2 ODD numbers is EVEN : $17 – 7 = 10$
- DIFF of ODD & EVEN numbers is ODD : $13 – 8 = 5$

MULTIPLICATION

- PRODUCT of 2 EVEN numbers is EVEN : 6x8=48
- PRODUCT of 2 ODD numbers is ODD : 3x17=51
- PRODUCT of ODD & EVEN numbers is EVEN: 3x6=18

LAST DIGIT is of the product is the same as the last digit of the product of the last digits of the two numbers.

- E.g. For 987**6** x 843**2** = 832773**2**

- No of Digits in the product cannot exceed the sum of the digits of the two numbers.

DIVISION

- DIVISION of ODD number by ODD is ODD : $21 \div 3 = 7$
- DIVISION of EVEN number by ODD is EVEN: $24 \div 3 = 8$
- DIVISION of EVEN number by EVEN is ODD/EVEN

:$12 \div 4 = 3$, $12 \div 6 = 2$

Sum of Natural Numbers

- **Rule 1 :** Sum of first n natural numbers = $\frac{n(n+1)}{2}$

e.g. sum of 1 to 74 =74 x (74+1)/2 = 2775.

- **Rule 2 :** Sum of first n odd numbers = n^2

e.g. sum of first seven odd numbers

= (1+3+5+7+9+11+13) = 49 = 7^2.

- **Rule 3** : Sum of first n even numbers = n (n+1)

e.g. sum of first 9 even numbers

= (2+4+6+8+10+12+14+16+18) = 90

= 9 (9+1) = 9 x10 = 90

Sum of Natural Numbers

- **Rule 4 :** Sum of squares of first n natural numbers = $\frac{n(n+1)(2n+1)}{6}$

e.g. sum of squares of first 8 natural numbers

= (1 + 4 + 9 + 16 + 25 + 36 + 49 + 64) = 204

= 8 (8+1)(16+1)/ 6 = 8 x 9 x 17 / 6 = 204

- **Rule 5 :** Sum of cubes of first n natural numbers = $[n(n+1)/ 2]^2$

e.g. sum of cubes of first 4 natural numbers

= (1 + 8 + 27 + 64) = 100

= $[4 (4+1)/2]^2$ = 100

DIVISION

- DIVISION by ZERO is NOT POSSIBLE
- If two numbers are divisible by a number then their sum & difference is also divisible by the number.
- E.g. For 63 is divisible by 9. 27 is also divisible by 9.
- So 63 + 27 = 90 is also divisible by 9
- And 63 – 27 = 36 is also divisible by 9

DIVISIBILITY RULES

- 2 : Unit place is even or zero(last digit should be divisible by 2)
- 3 : Sum of the digits is divisible by 3. e.g : 324
- 4 : Last 2 digits are divisible by 4 or last 2 digits are 0. e.g : 324
- 5 : Unit digit is 5 or 0
- **6 : Divisible by co primes 2 & 3.** e.g : 324
- 8 : Number formed by last 3 digits is divisible by 8 or last 3 digits are 0. e.g : 1088
- 9 : Sum of all digits is divisible by 9. e.g : 324
- 10: Units digit is 0.

DIVISIBILITY RULES

- 11 : Difference between sum of digits in odd & even places should either be zero or divisible by 11

e.g: 8283

e.g : 918071

- **12 : Divisible by co primes 3 & 4** e.g : 324
- **14 : Divisible by co primes 2 & 7**
- **15** : **Divisible by co primes 3 & 5**
- **16 :** No formed by last 4 digits divisible by 16/ last 4 digits 0.
- **18 : Divisible by co primes 2 & 9**
- 20 : Units digit 0 & tens digit is even.

DIVISIBILITY RULES

- 7 : The difference between the two alternate groups taking 3 digits at a time should either be zero or multiple of 7.

    eg- 550500006

    eg- 7370356

- 13 : The difference between the two alternate groups taking 3 digits at a time should either be zero or multiple of 13.

    eg- 200174

DIVISIBILITY RULES

- **17: A number is divisible by 17 if you multiply the last digit by 5 and subtract that from the rest. If that result is divisible by 17, then your number is divisible by 17.**
- For example, for 986, then : 98 - (6 x 5) = 68.
- Since, 68 is divisible by 17, then 986 is also divisible by 17.
- Also, 876 is not divisible by 17 because 87 - (6 x 5) = 57 and 57 is not divisible by 17.

- **19: To determine if a number is divisible by 19, take the last digit and multiply it by 2. Then add that to the rest of the number. If the result is divisible by 19, then the number is divisible by 19.**
- For example, 475 is divisible by 19 because 47 + (5 x 2) = 57, and 57 is divisible by 19.
- But , 575 is not divisible by 19 because 57 + (5 x 2) = 67, and 67 is not divisible by 19.

PROPERTIES OF DIVISIBILITY

To find a number completely divisible by another :

A) **Greatest 'n' digit number exactly divisible by a Number :**

Method : By subtracting the remainder

e.g a) Greatest 3 digit number divisible by 13

Greatest 3 digit number = 999. 999/13 gives remainder 11.

999-11 = 988 = Greatest 3 digit number divisible by 13

B) **Least 'n' digit number exactly divisible by a Number :**

Method : By adding the (divisor – remainder)

e.g b) Least 3 digit number divisible by 13

Least 3 digit number = 100. 100/13 gives remainder 9

100 + (13 – 9) = 104 = Least 3 digit number divisible by 13

PROPERTIES OF DIVISIBILITY

Q. On dividing a number by 999, the quotient is 366 and the remainder is 103. The number is:

A.364724 B.365387 C.365737 D.366757 E. None of these

Soln-

dividend = divisor x quotient + remainder

Required number = 999 x 366 + 103

= (1000 - 1) x 366 + 103

= 366000 - 366 + 103

= 365737

Ans: C

PROPERTIES OF DIVISIBILITY

Q. A number when divided by 342 gives remainder 47. If the same number is divided by 19 what would be the remainder?

A. 7 B. 8 C. 9 D. 10

Soln :

On dividing the number by 342 let the quotient be q.

→ Number = 342q + 47

→ Number = (19 x 18q) + (19x2 + 9)

→ Number = 19(18q+2) + 9

→ When number is divided by 19 it gives 18q+2 as the quotient & remainder = 9

Ans : C